Intro to AI — Project Ideas (backup list)

This is a list of **ready-to-use projects** you can pick from for your main project. It's mainly composed of projects from the previous years.

It is a backup, not the default. By default, you should define your own project, based on what your binôme wants to work on and your prior knowledge (see `main_project.pdf`). Use this list only if you cannot converge on your own idea in time, or as inspiration to spark one.

Each project comes with a **core task** (a reasonable minimum) and one or more **advanced extensions** (directions to go further). Dataset links are given where available; check size, format, and license against the criteria in `main_project.pdf` before committing.

Audio

Mood-Based Music Analysis & Composition

Create a system to analyze the emotional landscape of audio tracks.

- **Core task:** Classify songs by mood (e.g. joyful, melancholic) and analyze genre correlations.
- **Advanced extensions:** Implement instrument detection to identify dominant sounds and their role in mood setting. Optionally, include leading melody estimation.

Audio-Driven Playlist Generation

- **Core task:** Curating a custom playlist by analyzing the overall mood and style of multiple user-provided audio tracks.
- **Advanced extensions:** Adapt the playlist given text prompts, such as the mood of the listener, specific bands, genres or music, or more subjective evaluation ("I don't like this song", etc.).

Image

Sports Image Classification

Classify the sport displayed in an image.

- **Core task:** Classify different sports based on input images.
- **Advanced extensions:** Generate textual descriptions for sports, or use generative AI to create synthetic sports images and compare prompts to see which features are most useful for classification.

Geo-Localization ("GeoGuessr" Model)

A computer vision challenge to determine location from visual cues.

- **Core task:** Given an image, predict its geographic location.
- **Advanced extensions:** Interpret the decisions of the models, and propose a text description of the image related to the geolocalisation.

Museum Curation and Exhibition Design

- **Core task:** Given a set of paintings, organize them into coherent "rooms" and identify "bridge" pieces that create smooth transitions between artistic movements.
- **Advanced extension:** Semantically analyze the generated rooms and automatically produce thematic descriptions or titles for each room.

Text

Movie Recommendation Engine

- **Core task:** Develop a content-based recommendation system that analyzes movie plots to suggest films similar to those already viewed by the user (e.g. Wikipedia Movie Plots).
- **Advanced extension:** Implement a semantic recommendation feature based on user prompts. Create a new movie plot based on user prompts.

Fake News Classification & Emotional Analysis

- **Core task:** Distinguish real from fake news. The model must output not just a label, but also the probability or confidence score of an article being fake, enabling quantitative risk assessment.
- **Advanced extensions:** Conduct an emotion analysis (e.g. using specialized NLP models) on the text and investigate the correlation between specific emotional tones (e.g. fear, anger, surprise) and the classification of an article as fake news. Interpret the outputs of the model to identify the specific words or phrases that drive the model's 'fake' or 'real' classification.

French Football Specialized LLM

- **Core task:** Develop an LLM specialized in the French football league.
- **Advanced extensions:** Predict information about the future of the football leagues (future winners, transfers, etc.). Develop LLMs for different European/International football (or different sports) leagues.

Law and Bills Analysis

- **Core task:** Based on the explanatory data of the 2025 Finance Bill amendments, develop an LLM able to predict the likely political party responsible for drafting each amendment.
- **Advanced extension:** Extend to the authors. Interpret the decision of the model: why is this bill classified as this party? Generate new bills.

Spoiler Detection in User Reviews

- **Core task:** Analyze user reviews and accurately determine whether or not they contain spoilers (IMDB Spoiler Dataset).
- **Advanced extension:** Go beyond simple classification by implementing a spoiler localization model. The goal is to identify the specific sentences or segments of text that constitute the spoiler, enabling an automated system to redact or blur the sensitive information within the review.

Video

Soundless Video Captioning

- **Core task:** Caption videos using only the image, without the audio information (dataset: VISPER).
- **Advanced extensions:** Compare the results in different languages. Automatically generate the speech (audio) with an audio generation model. Generate the continuation of the video (with/without audio).

Multimodal

Multimodal YouTube Title Generation & Optimization

- **Core task:** Generate an efficient YouTube title based on the video's thumbnail image and a custom text prompt detailing the video's context and goals (e.g. target views, topic) (dataset: YouTube).
- **Advanced extension:** Develop a secondary evaluation model to score the generated titles against critical metrics: relevance to the content, originality, and predicted attractiveness/view maximization potential. This model could then be used to create a feedback loop to refine the initial title generation process.

Weather Classification & Multimodal Retrieval

- **Core task:** Identify four main weather conditions (Sunrise, Shine, Rain, Cloudy) and build a text-to-image retrieval system to find images based on descriptive queries.
- **Advanced extensions:** Retrieve an image from the dataset given a text prompt. Implement fine-grained classification to distinguish specific cloud sub-types within the 'Cloudy' class. Develop a multimodal manipulation feature (Image + Text input) to modify a scene based on textual instructions (e.g. "make this cloudy sky sunless").